in the history of humanity. See Steve Jones, *Revolutionary Science: Transformation and Turmoil in the Age of the Guillotine* (New York: Pegasus, 2017).

11. Changing what is opportune: for instance, the sign of the magnetic field in the equations of Maxwell, charge and parity of elementary particles, etc. It is the invariance under CPT (Charge, Parity, and Time reversal symmetry) that is relevant.

12. The equations of Newton determine how things *accelerate*, and the acceleration does not change if I project a film backward. The acceleration of a stone thrown upward is the same as that of a falling stone. If I imagine years running backward, the moon turns around the Earth in the opposite direction but appears equally attracted to the Earth.

13. The conclusion does not change by adding quantum gravity. On the efforts to fund the origin of the direction of time, see, for example, H. D. Zeh, *Die Physik der Zeitrichtung* (Berlin: Springer, 1984).

14. Rudolf Clausius, "*Über verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie*," *Annalen der Physik* 125 (1865): 353–400; 390.

15. In particular as a quantity of heat that escapes from a body *divided by temperature*. When the heat escapes from a hot body and enters a cold one, the total entropy increases because the difference in temperature makes it so that the entropy due to heat that escapes is less than that owed to the heat that enters. When all the bodies reach the same temperature, the entropy has reached its maximum: equilibrium has been reached.

16. Arnold Sommerfeld.

17. Wilhelm Ostwald.

18. The definition of entropy requires a *coarse graining*, that is to say, the distinction between microstates and macrostates. The entropy of a macrostate is determined by the number of corresponding microstates. In classic thermodynamics, the *coarse graining* is defined the moment it is decided to treat some variables of the system as "manipulable" or "measurable" from outside (the volume or pressure of a gas, for instance). A macrostate is determined by fixing *these* macroscopic variables.

19. That is to say, in a deterministic manner if you overlook quantum mechanics, and in a probabilistic manner if you take account of quantum mechanics instead. In both cases, in the same way for the future as for the past.

20. $S = k \log W$. S is the entropy, W is the number of microscopic states, or the corresponding volume of phase space, and k is just a constant, today called Boltzmann's constant, that adjusts the (arbitrary) dimensions.

3. THE END OF THE PRESENT

21. General relativity (A. Einstein, "Die Grundlage der algemeinen Relativitätstheorie," op. cit.).

22. Special relativity (Albert Einstein, "Zur Elektrodynamik bewegter Körper," *Annalen der Physik* 17 [1905]: 891–921).

23. J. C. Hafele and Richard E. Keating, "Around-the-World Atomic Clocks: Observed Relativistic Time Gains," *Science* 177 (1972): 168–70.

24. That depends as much on t as on your speed and position.

25. Poincaré. Lorentz had tried to give a physical interpretation to t, but in a quite convoluted way.